period of your orbit and plugging the results into Newton's formula, you determine the mass of the hole. It is 1 million times the mass of the Sun, just as claimed in Schechter's Black-Hole Atlas. From the absence of any tornado-like swirl in the inflowing gas and dust you infer that the hole is not spinning much; its horizon, therefore, must be spherical and its circumference must be 18.5 million kilometers, eight times larger than the Moon's orbit around the Earth.

After further scrutiny of the infalling gas, you prepare to descend toward the horizon. For safety, Kares sets up a laser communication link between your space capsule and your starship's master computer, DAWN. You then drop out of the belly of the starship, turn your capsule so its jets point in the direction of your circling orbital motion, and start blasting gently to slow your orbital motion and drive yourself into a gentle inward (downward) spiral from one coasting circular orbit to another.

All goes as expected until you reach an orbit of circumference 55 million kilometers—just three times the circumference of the horizon. There the gentle blast of your rocket engine, instead of driving you into a slightly tighter circular orbit, sends you into a suicidal plunge toward the horizon. In panic you rotate your capsule and blast with great force to move back up into an orbit just outside 55 million kilometers.

"What the hell went wrong!?" you ask DAWN by laser link.

"Tikhii, tikhii," she replies soothingly. "You planned your orbit using Newton's description of the laws of gravity. But Newton's description is only an approximation to the true gravitational laws that govern the Universe.[28] It is an excellent approximation far from the horizon, but bad near the horizon. Much more accurate is Einstein's general relativistic description; it agrees to enormous precision with the true laws of gravity near the horizon, and it predicts that, as you near the horizon, the pull of gravity becomes stronger than Newton ever expected. To remain in a circular orbit, with this strengthened gravity counterbalanced by the centrifugal force, you must strengthen your centrifugal force, which means you must increase your orbital speed around the black hole: As you descend through three horizon circumferences, you must rotate your capsule around and start blasting yourself forward. Because instead you kept blasting backward, slowing your motion, gravity overwhelmed your centrifugal force as you passed through three horizon circumferences, and hurled you inward."